Rooted: Hackathon Project Plan

A health and doomscrolling app that grows a virtual garden. Working title "Rooted". Plan for a 1.5-day hackathon, from idea to demo.

1. One-Line Pitch

"Forest rewards you for putting your phone down. Rooted rewards you for living well."

Rooted combines your wearable health data (steps, sleep) with your doomscrolling time. Healthy habits earn coins, coins buy plants and garden tools, and the garden thrives or goes dormant depending on how you actually live. Milestones also connect to a real garden or real tree planting.

2. Problem Statement

1. **Modern life pulls us indoors and onto screens.** Doomscrolling eats time, raises stress, and cuts into sleep and movement.
2. **Health data is fragmented.** Garmin, Apple Watch, and Oura each show different numbers in different apps. Aggregators exist (Vora, fitMetrics, 4Sight and others) but mostly display data without changing behavior.
3. **Focus apps are shallow.** Forest-style apps reward only one behavior (not touching the phone) and use a cartoon tree and a streak counter that people stop caring about.
4. **Nothing ties the digital habit to the natural world.** There is no link between your habits and a garden, or a real tree.

3. Objective

Main objective: Build a working app that turns daily healthy behavior (walking, sleeping well, reducing doomscrolling) into a living virtual garden, with a visible link to real-world planting.

Sub-objectives:

- Pull health data from at least one real or realistic source through a unified data layer.
- Measure doomscrolling time and compare it to the user's own baseline.
- Run a simple, fair reward economy (tasks, coins, shop, growth).
- Show a garden that visibly responds to behavior.
- Demonstrate the virtual-to-real link (real tree or community garden), even if the payment step is mocked.

Success metrics for the demo:

- A live demo completes the full loop in under 3 minutes: data in, task completed, coins earned, item bought, plant grows.
- A judge understands the problem and the twist in one sentence.
- The app works on a phone-sized screen without crashing.

4. Target User

People aged 18 to 35 who own a wearable or just a smartphone, feel they scroll too much, and want a gentle (not guilt-based) nudge toward sleep, movement, and time outside.

5. The Core Loop

1. **Sync:** Health data (steps, sleep) and screen/scroll data arrive.
2. **Evaluate:** The app checks the day's tasks against that data.
3. **Reward:** Completed tasks earn coins and "sunlight" (a garden-health boost).
4. **Spend:** Coins buy seeds, plants, flowers, a watering can, and fertilizer.
5. **Grow:** The garden grows based on care and real behavior. Neglect makes plants go dormant (they never die, to avoid guilt).
6. **Give back:** Milestones fund or point to a real tree or garden.

6. Feature List

MVP (must be done by the end of day 1.5)

- Sign-in (or a one-tap demo user).
- Daily task list with live progress bars: steps, sleep, doomscroll reduction.
- Data adapter layer with a mock/seeded source plus one real source (see section 8).
- Coin wallet and transaction history.
- Shop: seeds, flowers, trees, watering can, fertilizer.
- Virtual garden grid with plants that grow through stages.
- Garden health meter (7-day rolling).
- "Real impact" screen showing a real tree planted (mocked payment) with a certificate card.

Stretch (only if MVP is done and polished)

- Chrome extension that measures real doomscrolling time on social sites.
- Seasonal events and rare plants.
- Friends' shared garden.
- Community garden map.
- Push notifications and "outside break" suggestions.

Explicitly out of scope

Accounts with real payments, medical advice, native iOS/Android builds, supporting every wearable.

7. Tasks, Rewards, and the Game Economy

Keep the economy simple so it can be built and explained quickly. All numbers below are starting values to tune during testing.

7.1 Daily tasks

Task	Condition	Reward
First steps	1,000 steps	10 coins
Active day	5,000 steps	+20 coins
Strong day	8,000+ steps	+30 coins
Good sleep	7+ hours of sleep	+25 coins
Sleep consistency	Bedtime within 30 minutes of your target	+15 coins
Doomscroll down	Scroll time at least 20% below your 7-day average (or under a fixed cap on day 1)	+25 coins
Offline hour	1 hour with no distracting apps (up to 3 per day)	+15 coins each
Outside time (stretch)	20+ minutes outdoors	+20 coins

Fairness rule: doomscrolling goals compare against your own baseline, not a universal number. This keeps the game fair for light and heavy users alike.

7.2 Shop

Item	Type	Price	Effect
Daisy / Marigold seed	Flower	20 to 30	Blooms in 3 days of care
Sunflower seed	Flower	50	Blooms in 5 days
Fruit sapling	Tree	100	Matures in 7 days
Oak sapling	Tree	150	Matures in 10 days
Watering can	Tool (one-time)	50	Unlocks daily watering
Fertilizer	Consumable	20 each	Doubles growth for 24 hours on one plant
Rare seed (stretch)	Special	200	Local native species

7.3 Growth mechanics

- Each plant has **growth points** and stages: seed, sprout, young, mature, blooming.
- Each day, a plant gains growth when it is **watered** (needs the watering can) and the garden receives **sunlight**.
- **Sunlight** comes from your behavior: each completed task adds to a daily sunlight percentage (for example, 3 or more tasks equals full sunlight).
- **Fertilizer** doubles that plant's growth for 24 hours.
- **No death:** if you skip days, plants turn dormant (greyed out) and recover when you return. This keeps the tone encouraging, not punishing.
- **Garden health meter** is a 7-day rolling average of task completion, shown on the main screen.

8. Data Sources and Integrations

This is the riskiest part of the project, so decide early and always keep a fallback.

8.1 Health data (steps, sleep)

Each wearable has its own API, approval process, and quirks. Do not build four integrations in 1.5 days. Use a **data adapter layer**: the app only ever talks to a standard internal format, and each source is a plug-in behind it.

Standard internal format (daily summary):

```
{ date, steps, sleepMinutes, bedtime, wakeTime, activeMinutes, source }
```

Adapters, in order of priority:

1. **Mock adapter (required).** Seeded realistic data for 14 days plus a "live demo" control that lets you bump steps or sleep on stage. This guarantees the demo works.
2. **File import adapter (strongly recommended).** Import an Apple Health export or a CSV from Oura or Garmin. It uses real data with no API approval, so it is the best "it's real" proof.
3. **Aggregator API adapter (if time).** Services that unify many wearables behind one API (examples to check: Terra, Open Wearables). Verify current free tiers, sandbox access, and sign-up time at the start of the hackathon.
4. **Native adapters (post-hackathon).** Apple HealthKit requires an iOS app; Android's Health Connect requires an Android app. Garmin and Oura direct APIs typically require developer approval or OAuth setup.

Be upfront with judges: the app is wearable-agnostic by design, one source is live, and the rest plug into the same adapter.

8.2 Doomscrolling data

Measuring screen time on other apps is restricted on phones, so plan for this honestly.

- **Option A: Chrome extension (recommended stretch).** Manifest V3 extension that tracks time on a user-selected list of sites (for example, X, Instagram, TikTok, Reddit, YouTube) and flags long continuous sessions. It posts daily totals to your backend. This is real, doable in a few hours, and demo-friendly.
- **Option B: Manual / simulated input.** A slider or quick-log button plus seeded data. This is the guaranteed fallback.

- **Option C: Native screen-time APIs (post-hackathon).** iOS Screen Time API and Android usage stats need native apps and, on iOS, special entitlements. Mention this in the roadmap, do not attempt it now.

Doomscroll definition used by the app: total time on selected "feed" apps/sites per day, plus the number of sessions longer than 10 minutes. Rewards use total time compared to baseline.

8.3 Optional environment layer (strengthens the nature theme)

Add one free data source, such as Open-Meteo or AirNow, to show "today is a good day for a walk" (air quality, temperature, daylight). It is a small add-on that makes the nature link obvious.

9. Recommended Tech Stack

Chosen for speed, free tiers, and easy deployment.

Layer	Choice	Why
Frontend	Next.js (React) + Tailwind CSS	Fast to build, mobile-friendly, easy deploy
Garden visuals	React + SVG/CSS (or Phaser if someone knows it)	SVG is quick and animates easily
Charts	Recharts	Simple progress and trend charts
Backend / API	Next.js API routes (serverless)	One codebase, no separate server
Database + auth	Supabase (Postgres + auth) or Firebase	Free tier, instant setup
Data adapters	TypeScript modules in /adapters	Swappable sources
Doomscroll tracker	Chrome extension (Manifest V3)	Real browser data with no native app
Hosting	Vercel	One-click deploys, shareable demo link
Design	Figma (or sketches on paper)	30-minute wireframes, no more
Assets	Free icon/sprite packs (for example Kenney assets), emoji, or hand-drawn SVG	Avoid copyright problems and save time

Why a web app and not native: it ships in 1.5 days, runs on any phone browser, and avoids app-store and HealthKit complexity. State this as a deliberate scoping decision.

10. System Architecture

```
[Wearable export / mock data / aggregator API]
                |
        Data Adapter Layer  --->  Standard daily summary
                |
[Chrome extension / manual input] --> Scroll time
                |
          Backend API (Next.js routes)
   - Task evaluator  - Reward engine  - Garden engine
                |
        Database (Supabase)
                |
   Frontend: Dashboard | Tasks | Garden | Shop | Impact
                |
   Real-world link: tree-planting partner / community garden
```

Key logic modules:

- **Task evaluator:** reads today's summary, marks tasks complete, awards coins (idempotent, so a task can only pay out once per day).
- **Garden engine:** runs once per day (or on page load) to advance plant growth using sunlight and watering.
- **Reward engine:** writes coin transactions, keeping a full ledger.

11. Data Model

- **users:** id, name, bedtime_target, step_goal, created_at
- **daily_summaries:** id, user_id, date, steps, sleep_minutes, bedtime, wake_time, scroll_minutes, long_sessions, source
- **tasks:** id, key, title, condition_type, threshold, reward_coins
- **task_completions:** id, user_id, task_id, date, coins_awarded (unique on user, task, date)
- **wallet_transactions:** id, user_id, amount, reason, created_at
- **shop_items:** id, name, type, price, growth_days, sprite
- **inventory:** id, user_id, item_id, quantity
- **garden_plants:** id, user_id, item_id, grid_x, grid_y, growth_points, stage, is_dormant, planted_at

- **real_impact:** id, user_id, type (tree or garden), partner, status, certificate_url, created_at

12. API Endpoints

Endpoint	Purpose
POST /api/sync	Receive a daily summary from an adapter or the extension
GET /api/tasks/today	Today's tasks with live progress
POST /api/tasks/evaluate	Check completions and award coins
GET /api/wallet	Balance and history
GET /api/shop	Items for sale
POST /api/shop/buy	Purchase an item (checks balance)
GET /api/garden	Plants, positions, stages, health meter
POST /api/garden/plant	Place an owned seed on a grid cell
POST /api/garden/water	Water a plant (requires watering can)
POST /api/garden/fertilize	Use fertilizer on a plant
POST /api/impact/redeem	Trigger a real tree/garden contribution
POST /api/demo/boost	Demo-only: bump steps/sleep to show live rewards

13. Screens and User Flow

1. **Onboarding (2 steps):** pick a name, set step goal and target bedtime, choose data source (demo, file import, or connected).
2. **Home / Garden:** the garden grid, coin balance, garden health meter, today's sunlight, and a "what's next" prompt.
3. **Tasks:** today's tasks with progress bars (steps, sleep, scroll) and checkmarks when complete.

4. **Shop:** items with prices; buying adds to inventory.
5. **Inventory / Planting:** tap a seed, tap a grid cell to plant; tap a plant to water or fertilize.
6. **Insights (stretch):** weekly chart of steps, sleep, and scroll time, with one auto-generated insight (for example, "On days you scrolled less, you slept 40 minutes longer").
7. **Real Impact:** your real tree or garden contributions, certificate card, and the local garden map (stretch).

Design principles: calm, warm colors; big visuals; no guilt language; celebrate wins with small animations.

14. Connecting to a Real Garden

Choose one for the demo and mention the others as the roadmap.

1. **Real tree planting through a partner (recommended).** Every milestone (for example, every 500 coins earned or a 14-day streak) triggers a real planting. Check what tree-planting services currently offer in terms of APIs and pricing (examples: Ecologi, Tree-Nation, One Tree Planted). For the hackathon, mock the payment step: show the confirmation and a certificate card with a location and date. Be transparent that it is a prototype.
2. **Community garden map.** Use OpenStreetMap data (community gardens tagged as landuse=allotments or leisure=garden) to show the nearest community garden and suggest a volunteer visit as a reward activity.
3. **Home garden sync (hardware, post-hackathon).** A soil moisture sensor (for example, an ESP32 with a capacitive sensor) or smart planter feeds the app, and your virtual plant mirrors your real one.
4. **Citizen science (post-hackathon).** Identify a real local plant or bird with iNaturalist or Merlin to unlock that species in your virtual garden.
5. **Physical rewards (future).** Native seed packets mailed at milestones.

15. Build Plan: 1.5 Days

Assume a team of 3 to 4. Adjust roles to your skills.

Roles

- **Frontend / UI lead:** screens, garden visuals, animations.
- **Backend / data lead:** API, database, task and reward logic.
- **Integration lead:** data adapters, file import, Chrome extension.
- **Pitch / design lead:** wireframes, assets, copy, demo script, slides.

Hour-by-hour timeline

Phase 0: Setup (hours 0 to 2)

- Agree on the scope in section 6 and lock the MVP.
- Create the repo, Vercel project, and Supabase project; share keys.
- Sketch wireframes for the 5 core screens on paper or Figma.
- Decide the data adapter plan (mock plus file import, aggregator only if sign-up is instant).

Phase 1: Foundations (hours 2 to 8)

- Backend: tables, seed data, `/api/sync`, `/api/tasks/today`, `/api/wallet`.
- Frontend: app shell, navigation, task screen with progress bars using seeded data.
- Integration: mock adapter and the standard summary format.
- Design: pick a palette, collect or draw plant sprites for 5 stages and 4 plants.

Phase 2: Core loop (hours 8 to 16)

- Task evaluator and reward engine (idempotent payouts).
- Shop, buy flow, and inventory.
- Garden grid, planting, watering, fertilizing, and the daily growth engine.
- Garden health meter and sunlight.
- *Checkpoint at hour 16: you can complete a task, earn coins, buy a seed, plant it, and see it grow. If not, cut stretch features immediately.*

Sleep (hours 16 to 22, do not skip this) Tired teams ship buggy demos. Rest.

Phase 3: Real data and impact (hours 22 to 30)

- File import adapter (Apple Health or CSV) and, if time, the aggregator API.
- Chrome extension (or manual doomscroll input as a fallback).

- Real Impact screen with a mocked tree-planting flow and certificate card.
- Open-Meteo layer (optional) for a "good day to walk" banner.

Phase 4: Polish and demo prep (hours 30 to 36)

- Animations for growth, coin earning, and planting.
- Fix bugs; test on a phone; test with a poor connection.
- Add the demo boost button.
- Record a backup demo video.
- Write the pitch and rehearse at least twice.

Final hour: feature freeze. Only fix crashes.

16. Testing Checklist

- A task cannot pay out twice for the same day.
- Buying with insufficient coins fails gracefully.
- Plants advance correctly with and without watering and sunlight.
- Dormant plants recover instead of disappearing.
- Empty states work (new user, no data, no plants).
- The app works on a phone-sized screen.
- The demo works with Wi-Fi off (use the backup video as insurance).
- Time-based logic can be simulated, because you cannot wait days during judging (add a "skip a day" demo control).

17. Demo Script (3 minutes)

1. **Hook (20 seconds):** "We all scroll too much and move too little. Wearables show us numbers, but nothing makes us act. Rooted turns healthy habits into a living garden."
2. **Show the problem (20 seconds):** Quick view of the scattered health data and a scroll-time number.
3. **Show the data (30 seconds):** Import or sync real data into one place.

4. **Complete a task live (40 seconds):** Use the demo boost to hit 1,000 steps; watch coins and sunlight rise.
5. **Spend and grow (40 seconds):** Buy a seed, plant it, water it; use "skip a day" to show growth.
6. **Real-world link (20 seconds):** Show the real tree certificate: "Every milestone plants a real tree."
7. **Close (10 seconds):** "Forest rewards you for putting your phone down. We reward you for living well."

Judging alignment: relatable problem in one sentence; live working demo; clear data sources; unique twist (multi-behavior reward plus real impact); honest scoping.

18. Risks and Fallbacks

Risk	Fallback
Wearable API approval or sign-up is slow	Use file import and the mock adapter, and say so
Chrome extension takes too long	Manual scroll-time input plus seeded data
Scroll tracking is inaccurate	Frame it as "time on feed sites" and let users pick sites
Garden art takes too long	Use emoji and simple SVG shapes with growth-stage scaling
Real tree partner API is unavailable	Mock the flow and label it a prototype
Live demo failure	Backup video and a seeded demo account
Scope creep	Section 6 is the contract; stretch features only after the hour-16 checkpoint

19. Privacy, Safety, and Ethics

- Health and browsing data are sensitive. Collect the minimum, store daily summaries only (not raw browsing history), and let users delete their data.

- State clearly that Rooted is not medical advice.
- Avoid shame mechanics: no plant deaths, no "you failed" messages; use encouraging copy.
- Avoid rewarding unhealthy extremes (for example, no unlimited step rewards and no sleep-deprivation incentives; cap daily coins).
- If you mention real tree planting, be transparent about how it works and who the partner is.
- Use only assets you have the right to use.

20. After the Hackathon (Roadmap)

1. **Native mobile app** with HealthKit and Health Connect for true background sync.
2. **Native screen-time integration** for iOS and Android.
3. **Conflict resolution:** reconcile disagreeing devices into one trusted sleep and recovery score and explain why they differ.
4. **Health meets environment insights:** correlate wellness with air quality, pollen, heat, and time in green space.
5. **Real garden sync:** soil sensors and smart planters.
6. **Community features:** shared family or friend gardens and local volunteer events.
7. **Business model ideas:** partnerships with tree-planting and gardening brands, employer wellness programs, and premium plants.

21. Final Pre-Submission Checklist

- [ ] MVP loop works end to end on a phone
- [ ] Deployed link works and is shared with the team
- [ ] Seeded demo account with a good-looking garden
- [ ] Demo boost and skip-a-day controls work
- [ ] Backup demo video recorded
- [ ] README with setup steps, tech stack, and data-source notes
- [ ] Slides: problem, solution, demo, tech, impact, roadmap
- [ ] Pitch rehearsed with a timer

- [ ] All third-party APIs, assets, and datasets credited
- [ ] Team knows who answers which judge question (technical, design, impact)